Arab Academy for Science, Technology & Maritime Transport

College of Artificial Intelligence ———— Alamein Campus.

Course	DS414	Information Retrieval and Search Engines
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1. Overview

This work proposes a **task-oriented multimodal information retrieval (IR) agent** that processes text, image, and audio inputs to retrieve relevant information and execute goal-driven tasks. Unlike traditional IR systems, the proposed framework integrates multimodal fusion, task planning, retrieval grounding, and tool execution into a unified architecture.

2. Multimodal Processing

Each modality is processed independently:

- **Text Encoder** produces contextual embeddings.
- **Image Encoder** extracts aligned visual-semantic representations.
- **Audio Encoder** converts speech to text and generates embeddings.

These embeddings are forwarded to the fusion layer.

3. Multimodal Fusion

We implement and compare:

- Concatenation fusion
- Cross-modal attention fusion
- Hierarchical fusion

The output is a unified multimodal representation ZmmZ_{mm}Zmm, enabling joint reasoning across modalities.

4. Task Understanding & Planning

The fused representation is passed to:

- A **task classifier** for intent detection

- A **planner** that decomposes tasks into structured action sequences
- A **memory module** to maintain context

This transforms task completion into a decision-making process rather than single-step inference.

5. Retrieval-Augmented Reasoning

The system integrates a multimodal retriever that:

1. Reformulates queries from ZmmZ_{mm}Zmm
2. Retrieves relevant documents or media
3. Re-ranks results using cross-modal similarity

This improves factual grounding and reduces hallucination.

6. Tool-Augmented Execution

For multi-step tasks, the planner invokes external tools (e.g., APIs, calculators, databases). Tool augmentation increases success rate on complex tasks and aligns with our hypothesis on tool-assisted reasoning.

7. Experimental Setup

- Image-text datasets: MS COCO, VQA
- Audio datasets: LibriSpeech
- Task dialogue datasets: MultiWOZ

We conduct ablation studies on fusion methods, retrieval integration, and tool usage.

Evaluation Criteria

To evaluate the system holistically, we define four metric groups:

1. Retrieval Metrics

- Precision
- Recall
- F1-score
- Cross-modal relevance
- OSWorld
- Llava-RE

2. Task Performance

- Task Success Rate
- Multi-step Accuracy

3. Efficiency

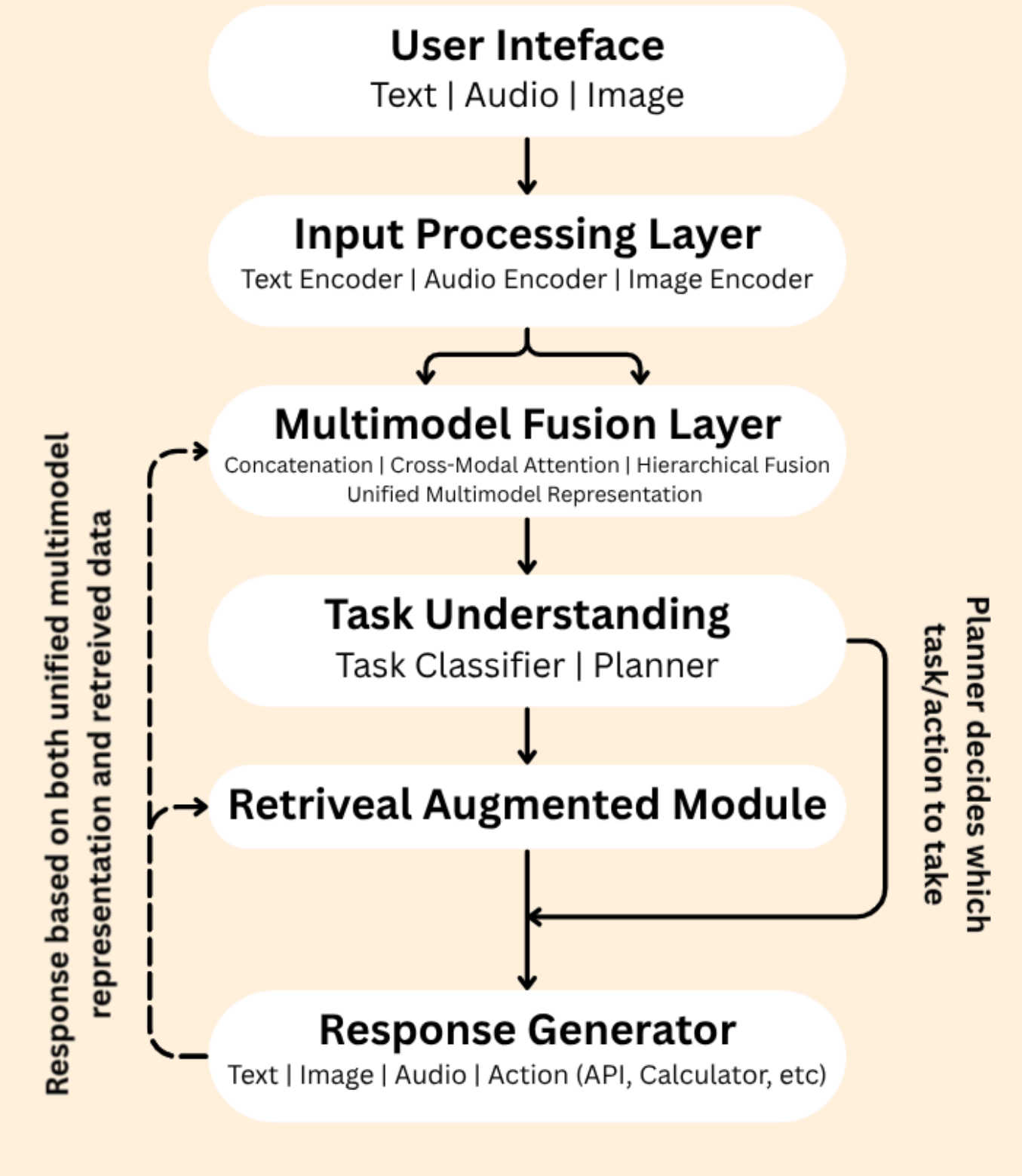
- Task Completion Time
- Number of Planning Steps

4. Fusion Impact

- Performance gain over unimodal baseline
- Ablation study comparisons

Composite Score

Final evaluation combines:

- 40% Task Success
- 30% Retrieval Quality
- 20% Efficiency
- 10% Cross-modal Relevance

System Framework